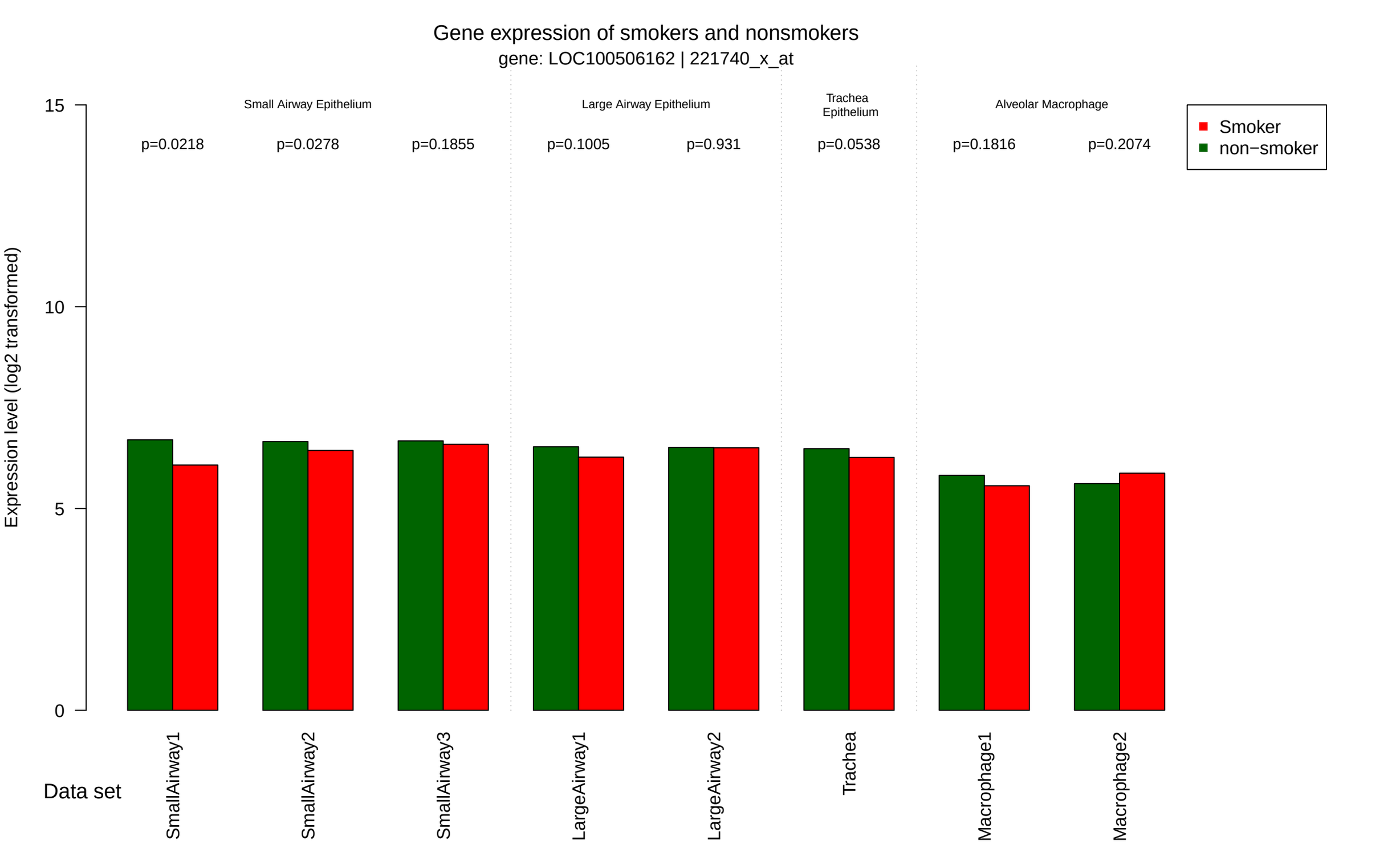
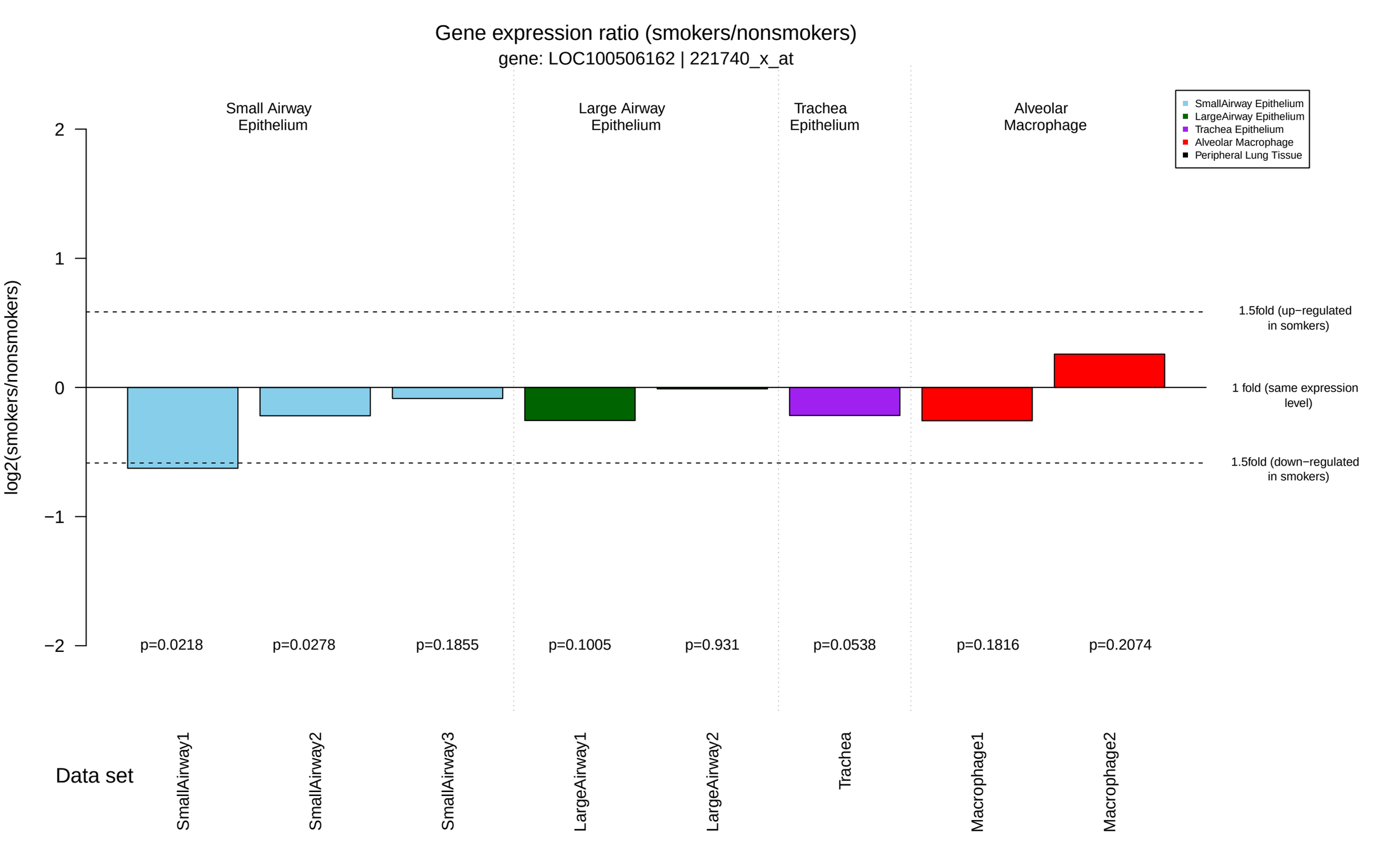
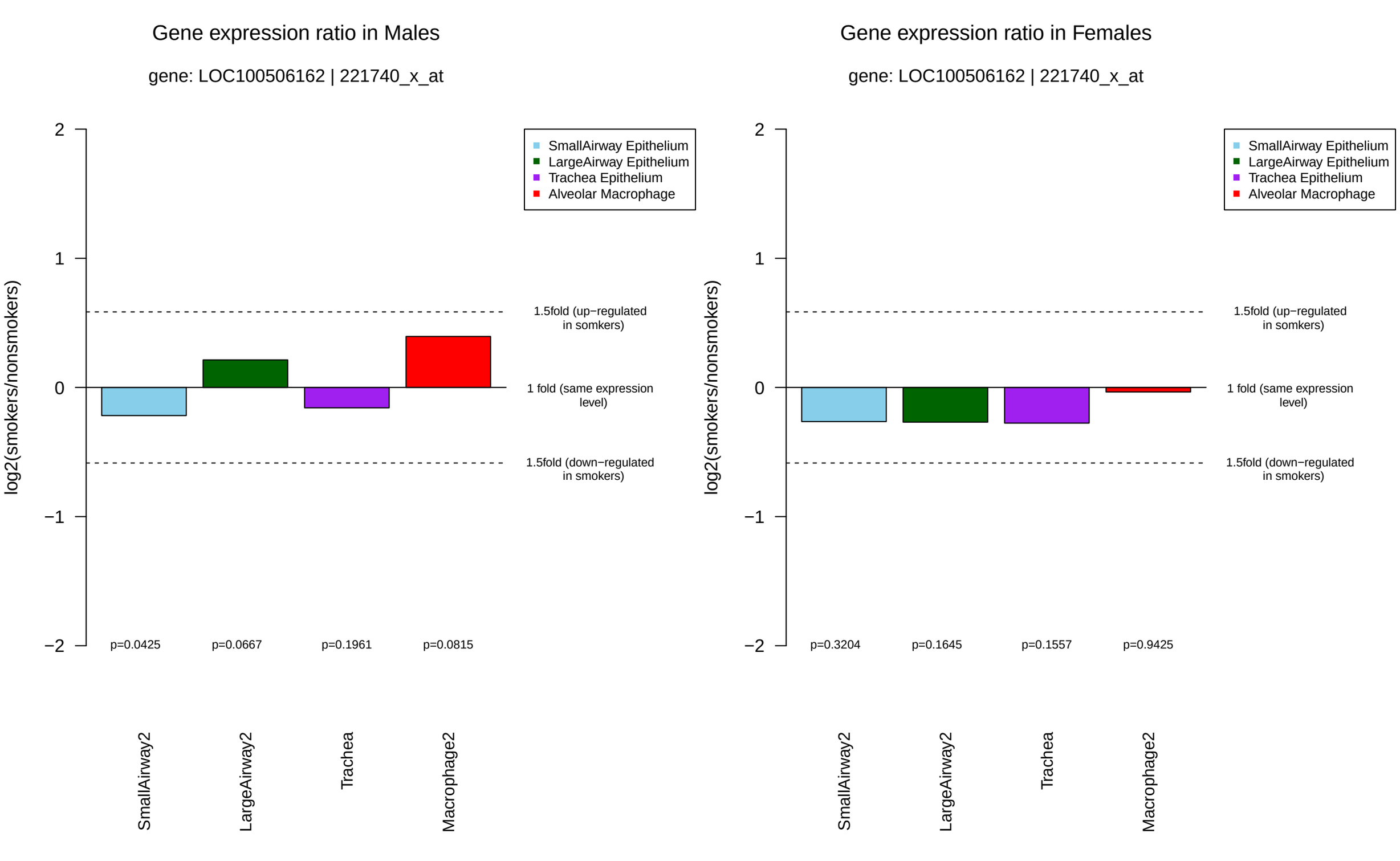
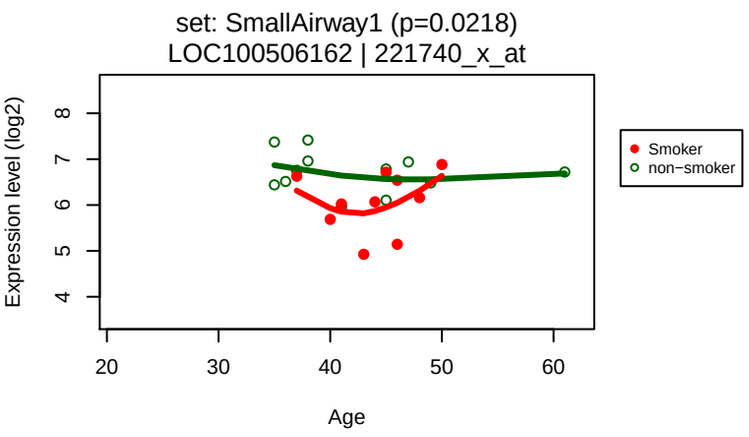
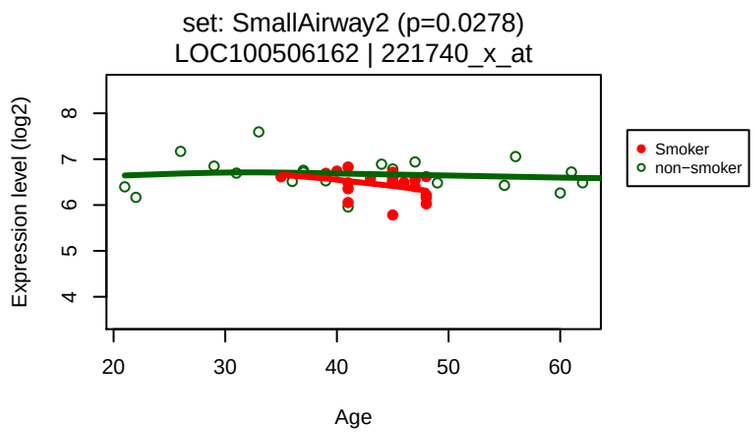
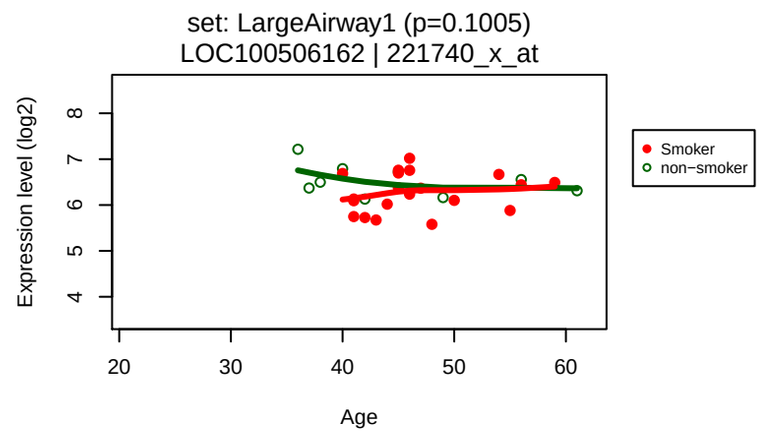

Gene expression of smokers and nonsmokers
gene: LOC100506162 | 221740_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0218
p=0.0278
p=0.1855
p=0.1005
p=0.931
p=0.0538
p=0.1816
p=0.2074
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

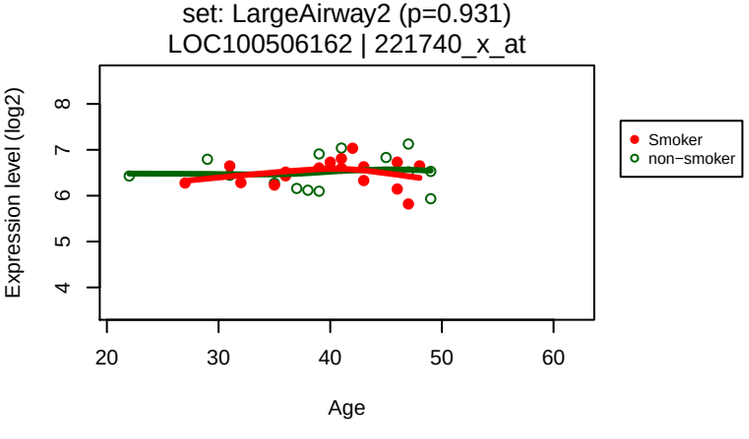

Gene expression ratio (smokers/nonsmokers)
gene: LOC100506162 | 221740_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0218
p=0.0278
p=0.1855
p=0.1005
p=0.931
p=0.0538
p=0.1816
p=0.2074
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

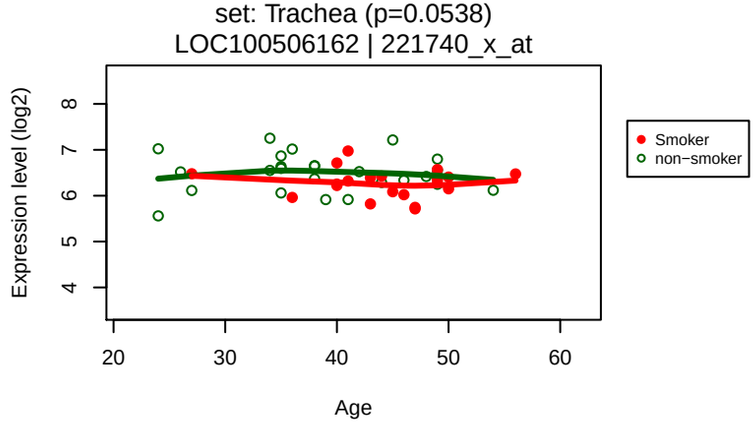

Gene expression ratio in Males
gene: LOC100506162 | 221740_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0425
p=0.0667
p=0.1961
p=0.0815
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC100506162 | 221740_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3204
p=0.1645
p=0.1557
p=0.9425
SmallAirway2
LargeAirway2
Trachea
Macrophage2

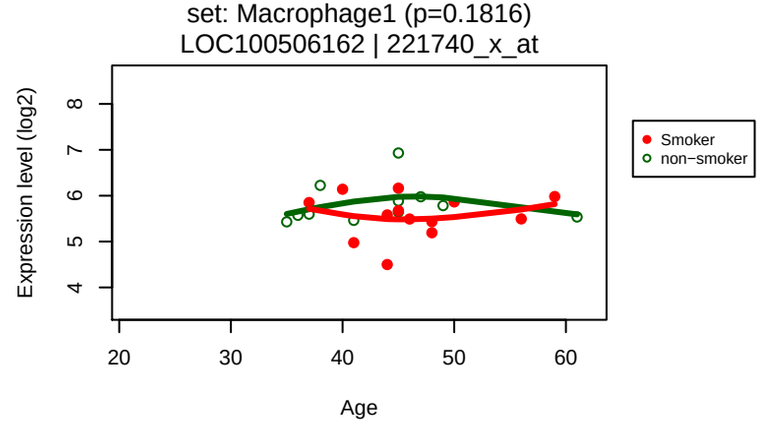

set: SmallAirway1 (p=0.0218)
LOC100506162 | 221740_x_at
Expression level (log2)
Age
Smoker
non−smoker

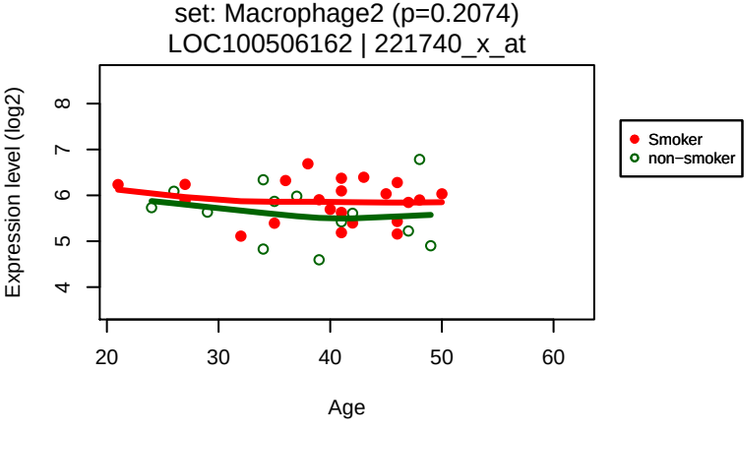

set: SmallAirway2 (p=0.0278)
LOC100506162 | 221740_x_at
Expression level (log2)
Age
Smoker
non−smoker

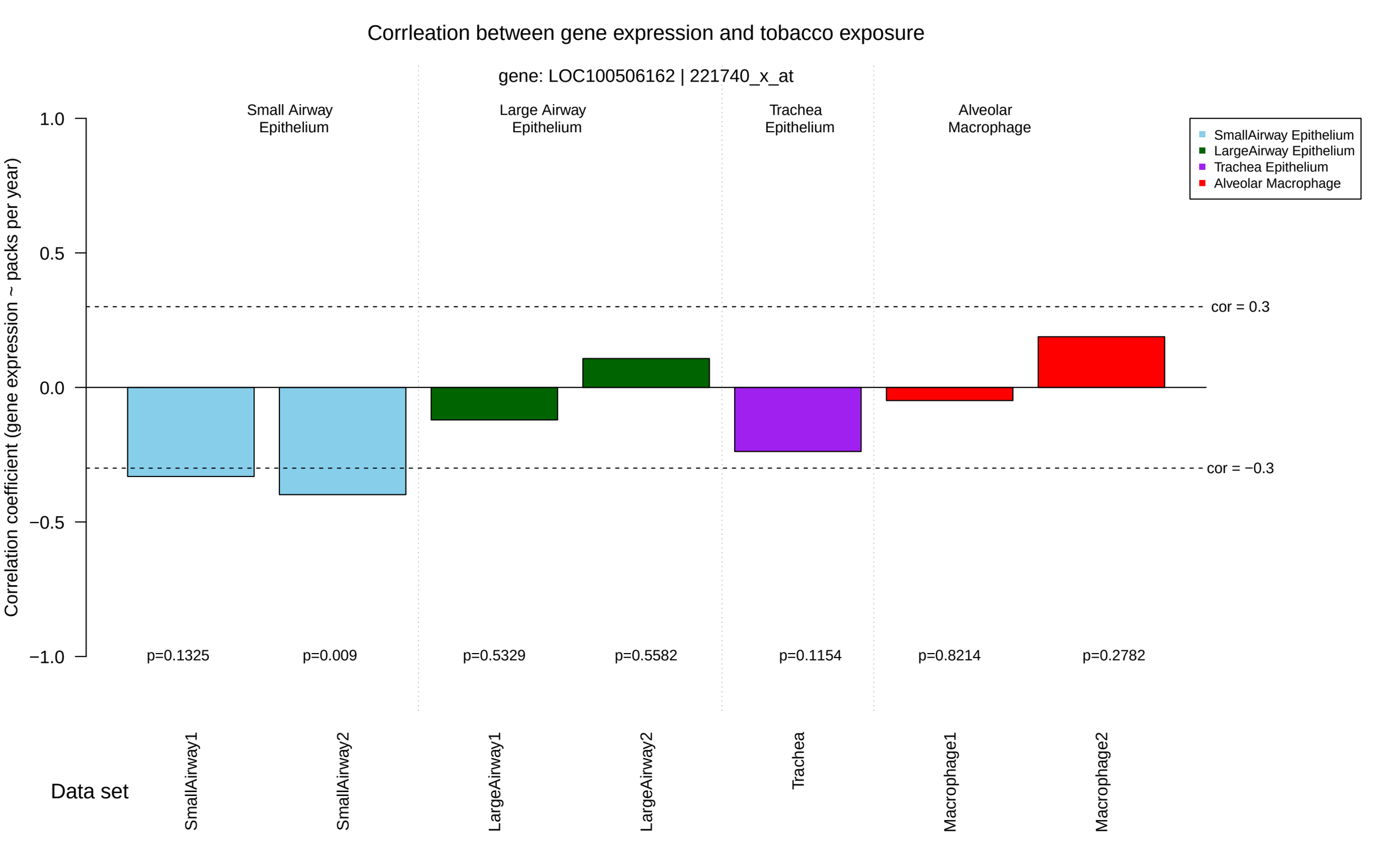

set: LargeAirway1 (p=0.1005)
LOC100506162 | 221740_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.931)
LOC100506162 | 221740_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0538)
LOC100506162 | 221740_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1816)
LOC100506162 | 221740_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2074)
LOC100506162 | 221740_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC100506162 | 221740_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1325
p=0.009
p=0.5329
p=0.5582
p=0.1154
p=0.8214
p=0.2782
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2